

A Walkway Through The Rainforest: Warm-Up

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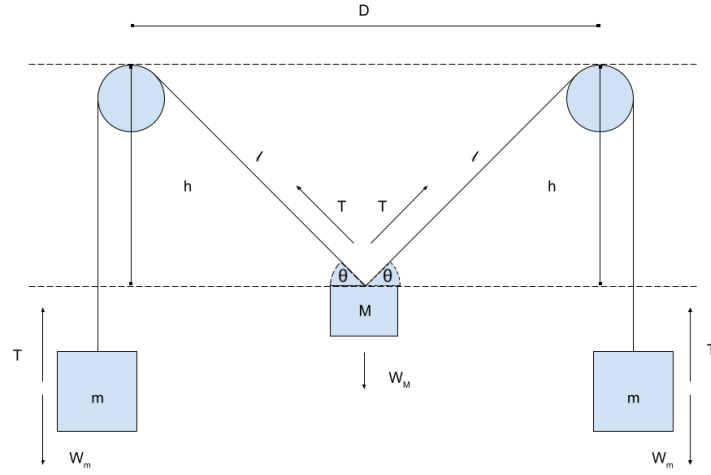


Figure 1: Diagram of the lab setup, with forces and important distances labeled

We want to find how the distance the walkway sags as a function of the masses in the problem, and the separation distance between the pulleys.

D M m
The first thing we want to do is set up Newton's second law for the central mass

$$\begin{aligned}
 M : \\
 x : F_{NET,x} &= T \cos \theta - T \cos \theta = Ma_{M,x} & (1) \\
 y : F_{NET,y} &= 2T \sin \theta - Mg = Ma_{M,y} & (2)
 \end{aligned}$$

Second, we look at the y equation for one of the side masses (we know it will be the same for both side masses)

$$m : \\ x : F_{NET,y} = T - mg = ma_{m,y}$$

Recall that $a_{m,y}$ is 0 because we are working in statics.

$$T = mg \tag{3}$$

Note that the x equation simplifies to $0 = a_x$ which makes sense: this is a static problem, and nothing is supposed to be moving anymore.

For the y equation (2), we can solve for the case where $a_y = 0$, again because we are solving a static problem.

$$2T \sin \theta = Mg$$

We can substitute (3) into T to get an expression in terms of m

$$2mg \sin \theta = Mg$$

thus

$$\begin{aligned} \sin \theta &= \frac{Mg}{2mg} \\ \sin \theta &= \frac{M}{2m} \end{aligned} \tag{4}$$

Also, recall that

$$\sin \theta = \frac{O}{H}$$

From the Pythagorean Theorem, we know

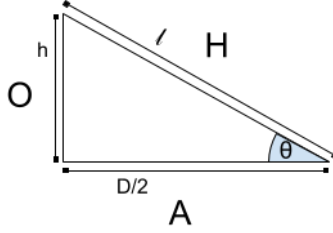
$$\ell = \sqrt{\frac{D^2}{4} + h^2}$$

And so

$$\sin \theta = \frac{O}{H} = \frac{h}{\sqrt{\frac{D^2}{4} + h^2}}$$

We can plug this into (4)

$$\sin \theta = \frac{M}{2m}$$



$$\frac{h}{\sqrt{\frac{D^2}{4} + h^2}} = \frac{M}{2m}$$

$$h = \frac{M}{2m} \sqrt{\frac{D^2}{4} + h^2}$$

We can square both sides

$$h^2 = \frac{M^2}{4m^2} \left(\frac{D^2}{4} + h^2 \right)$$

$$h^2 = \frac{M^2}{4m^2} \frac{D^2}{4} + \frac{M^2}{4m^2} h^2$$

$$h^2 - \frac{M^2}{4m^2} h^2 = \frac{M^2}{4m^2} \frac{D^2}{4}$$

$$h^2 \left(1 - \frac{M^2}{4m^2} \right) = \frac{M^2}{4m^2} \frac{D^2}{4}$$

$$h^2 = \frac{\frac{M^2}{4m^2} \frac{D^2}{4}}{\left(1 - \frac{M^2}{4m^2} \right)}$$

$$h^2 = \frac{\frac{4m^2}{M^2} \frac{M^2}{4m^2} \frac{D^2}{4}}{\frac{M^2}{M^2} \left(1 - \frac{M^2}{4m^2} \right)}$$

$$h^2 = \frac{\frac{D^2}{4}}{\left(\frac{M^2}{4m^2} - 1 \right)}$$

$$h = \sqrt{\frac{\frac{D^2}{4}}{\left(\frac{M^2}{4m^2} - 1 \right)}} \quad (5)$$